

PROJECT BOATY

# Design FMEA

Failure modes and effects analysis of the Mk1 architecture, used to drive simulator, rig and bench tests

|          |                                                                                                                                                                                                                   |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document | BOATY-FMEA-001                                                                                                                                                                                                    |
| Issue    | Issue G (baselined, sdr-baseline-1)                                                                                                                                                                               |
| Date     | 30 September 2026                                                                                                                                                                                                 |
| Status   | Baselined at the System Design Review (tag sdr-baseline-1); changes only by change request. Design stage (before hardware); updated after rigs, pool and before the first lake trial (SAF-006) by change request. |
| Basis    | ADD Issue H, ICD Issue H, SSS (HUL B, PRP C, PWR D, HLM F, MCP F, MCN F, REC B, SIM G), simulator evidence to the SDR                                                                                             |
| Content  | 60 failure modes, 29 actions, 72 catalogued tests                                                                                                                                                                 |

## Document control

| Issue               | Date                      | Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | By                                     |
|---------------------|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| A                   | 28 September 2026         | First issue: design FMEA at architecture level.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Claude (drafted)                       |
| B                   | 28 September 2026         | Actions carried into SSS-HLM/MCP/MCN Issue B, ICD Issue C and ADD Issue D; status and post-action ratings added. A-05, A-09, A-17 await the operations manual.                                                                                                                                                                                                                                                                                                                                                         | Claude, owner request                  |
| C                   | 28 September 2026         | CR-03: salvaged cells (FM-23 O 1 → 2; A-20 cell acceptance) and adult PIN (new FM-48). Operations manual BOATY-OPS-001 closes A-05, A-09, A-11, A-17, A-19.                                                                                                                                                                                                                                                                                                                                                            | Claude, owner decision                 |
| D                   | 29 September 2026         | Simulator evidence (software/results): FM-08 re-rated with its real cause (KCL); FM-49 to FM-53 added; actions A-21 to A-24 incorporated; A-18 closed (V-14: no native mechanism); detection re-rated where a scenario now passes.                                                                                                                                                                                                                                                                                     | Claude, owner decisions (CR-04, CR-05) |
| E                   | 29 September 2026         | Mission Control in the simulator (slice 3): FM-54 to FM-58 added from what the runs found, actions A-25 to A-29 incorporated; FM-09, 10, 34, 35, 41 re-rated now SC-24, 25, 31, 32 and 37 pass; FM-37 re-rated after the live Claude evaluation (SC-33).                                                                                                                                                                                                                                                               | Claude                                 |
| F                   | 30 September 2026         | SDR decision CR-07: FM-44 controls now include the 15 m nest stand-off enforced by the site linter (MCN-D65); SC-26 added to its tests. SDR WP2: FM-13 re-rated (O 3 → 2, D 3 → 2) now path planning is configured and SC-27 passes; FM-59 (RTL stalls near a zone) added from what SC-27 found. WP3: A-30 (B6 position-jump HOLD) after SC-20 found a sustained GNSS offset leaves the fence; FM-02 after-action RPN 108 → 72; FM-60 (home reset by re-arming on the water); S ≥ 9 rule text matches the build check. | Claude, owner decision                 |
| G                   | 30 September 2026         | SDR baseline candidate (WP5, RID-08): basis and references from the baseline register.                                                                                                                                                                                                                                                                                                                                                                                                                                 | Claude                                 |
| Role                | Name                      | Signature / date                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                        |
| Author              | Claude, for Project Boaty |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                        |
| Reviewer / approver | Project owner             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                        |

Review guidance: challenge the ratings, especially occurrence, which is a judgement before any hardware exists. And look for missing failure modes: an FMEA is only as good as its imagination.

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# 1. Purpose and scope

This design FMEA looks for ways Boaty Mk1 can fail and asks three questions of each. How bad is the effect? How likely is it? Will our planned tests find it before the boat goes on the lake? It works at the level of the ADD's subsystems and interfaces, before detailed design. It has two jobs:

- **Drive the tests.** Every failure mode with severity  $\geq 8$  must be exercised by a simulator, rig or bench test in the SSS-SIM catalogue. The build checks this.
- **Change the design early.** Where a risk is high, actions go into the next subsystem-spec issues while changes are still cheap.

It supports SAF-006 (FMEA before the first lake trial) and FS-013 (no single failure leaves the fence under power). It is a living document: ratings are revisited with rig, pool and lake evidence.

# 2. Method

## 2.1 Rating scales

| S  | Severity (effect)                                                               | O | Occurrence                       | D  | Detection by planned verification                  |
|----|---------------------------------------------------------------------------------|---|----------------------------------|----|----------------------------------------------------|
| 10 | Injury to a person, or fire                                                     | 8 | Expected most sessions           | 2  | Certain: an automated test exercises it directly   |
| 9  | Boat leaves the inclusion fence under power, or risk of collision with a person | 6 | Frequent: several times a season | 3  | High: a scripted scenario or rig test targets it   |
| 8  | Boat unrecoverable, or wildlife harmed or disturbed                             | 5 | Occasional                       | 5  | Moderate: inspection, analysis or an indirect test |
| 7  | Boat stops or drifts inside the fence; manual recovery needed                   | 4 | Likely during Mk1's life         | 7  | Low: likely to show only in pool or lake use       |
| 6  | Uncommanded behaviour, contained; boat returns                                  | 3 | Possible, but unlikely           | 9  | Very low: unlikely to be found before use          |
| 5  | Mission aborted; boat returns home safely                                       | 2 | Remote                           | 10 | None                                               |
| 4  | Mission degraded (no photos, no plan → templates)                               | 1 | Practically nil                  |    |                                                    |
| 3  | Minor inconvenience (reboot, re-plan)                                           |   |                                  |    |                                                    |
| 2  | Cosmetic or record gap                                                          |   |                                  |    |                                                    |
| 1  | No effect                                                                       |   |                                  |    |                                                    |

## 2.2 Rules

- Action required if  $S \geq 9$ , whatever O and D are, unless a direct test verifies the control ( $D \leq 3$ ): then the test stands in for the action (the rule the build checks; SDR RID-13)
- Action required if  $RPN = S \times O \times D \geq 100$
- Action required if  $D \geq 7$  and  $S \geq 7$
- Every row with  $S \geq 8$  must have at least one test

D rates our verification (will we find it before the lake?), not detection in operation. In-operation detection is listed separately as a control.

### 3. FMEA worksheet

Blue S cell = severity  $\geq 9$ . Orange RPN cell = at or above the action threshold. 'After' = RPN once the incorporated actions are in (Issue B); '-' = unchanged. Test IDs are defined in the SSS-SIM test catalogue (SSS-SIM Issue G).

| ID                  | Item / function          | Failure mode                            | Effect                                                            | S        | Cause                                                                                                    | O | Controls: prevent / detect in use                                                               | D | RPN        | After     | Tests               | Actions    |
|---------------------|--------------------------|-----------------------------------------|-------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------|---|------------|-----------|---------------------|------------|
| <b>FM-01</b><br>HLM | GNSS position            | No fix                                  | Motors stop (FS-004); boat drifts inside the fence                | <b>7</b> | Sky blocked, module or cable fault                                                                       | 3 | Mast mounting; M10 multi-constellation / EKF failsafe                                           | 3 | <b>63</b>  | -         | SC-04, L2-03        | -          |
| <b>FM-02</b><br>HLM | GNSS position            | Position jump (glitch)                  | False breach, or wrong steering near the fence                    | <b>9</b> | Multipath off water, reflections                                                                         | 4 | EKF glitch rejection; B6 position-jump HOLD (MCP-D35) / EKF innovation checks; B6 jump detector | 5 | <b>180</b> | <b>72</b> | SC-20, SC-13        | A-01, A-30 |
| <b>FM-03</b><br>HLM | GNSS position            | Frozen but reported valid               | Helm navigates blind; could leave the fence                       | <b>9</b> | Receiver hang, data line fault                                                                           | 2 | GNSS data-age checks / EKF failsafe                                                             | 5 | <b>90</b>  | <b>54</b> | SC-21, L2-04        | A-01       |
| <b>FM-04</b><br>HLM | Heading                  | Compass disturbed by motor current      | Curved tracks or circling                                         | <b>6</b> | Current loops near the compass                                                                           | 6 | Mast mount $\geq 150$ mm; interference calibration / EKF yaw checks; GNSS-velocity yaw fallback | 3 | <b>108</b> | <b>72</b> | SC-22, L2-05        | A-02       |
| <b>FM-05</b><br>HLM | Heading                  | Compass orientation wrong               | Boat drives the wrong way; RTL also wrong; persistent breach      | <b>9</b> | Parameter or mounting error                                                                              | 3 | Parameter baseline check / EKF consistency once moving                                          | 3 | <b>81</b>  | <b>36</b> | SC-22, SC-38, L2-06 | A-03       |
| <b>FM-06</b><br>HLM | State estimate           | EKF diverges                            | Erratic control; failsafe HOLD                                    | <b>7</b> | Sensor fault, vibration                                                                                  | 2 | Stable firmware, mounting / EKF failsafe                                                        | 3 | <b>42</b>  | -         | SC-04               | -          |
| <b>FM-07</b><br>HLM | Flight controller        | Firmware crash / reboot mid-mission     | Outputs stop; boots disarmed; drifts                              | <b>7</b> | Firmware fault                                                                                           | 2 | Pinned stable release / Heartbeat loss alarm                                                    | 5 | <b>70</b>  | -         | SC-23, L2-07        | -          |
| <b>FM-08</b><br>HLM | Flight controller supply | Brownout on a thrust step or near empty | Reboot as FM-07, during the very RTL the battery failsafe started | <b>7</b> | Pack sag below the FC's minimum input (Issue D: the F405-TE's 9 V floor, KCL KF-01; O 3 $\rightarrow$ 5) | 5 | PWR-D10/D12 / Logs                                                                              | 3 | <b>105</b> | <b>21</b> | L2-08, L2-09        | A-21       |

| ID                  | Item / function | Failure mode                                        | Effect                                         | S | Cause                                                                 | O | Controls: prevent / detect in use                                                                       | D | RPN | After | Tests        | Actions |
|---------------------|-----------------|-----------------------------------------------------|------------------------------------------------|---|-----------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------------------|---|-----|-------|--------------|---------|
| <b>FM-09</b><br>HLM | Configuration   | Parameters differ from baseline                     | A failsafe doesn't fire as specified           | 9 | Stale or edited parameters                                            | 4 | Baseline file (SAF-007) / Mismatch blocks arming (MCN-D45)                                              | 2 | 72  | 36    | SC-24        | -       |
| <b>FM-10</b><br>HLM | Fence           | Fence not loaded or disabled                        | No fence protection                            | 9 | Upload skipped or failed                                              | 3 | Pre-arm fence check / Arming refused                                                                    | 2 | 54  | 27    | SC-25        | -       |
| <b>FM-11</b><br>MCN | Site data       | Coordinates swapped or wrong site                   | Fence in the wrong place                       | 9 | GeoJSON [lon, lat] order error, file mix-up                           | 3 | Site linter / Pre-arm: home inside fence                                                                | 3 | 81  | 54    | SC-26        | A-04    |
| <b>FM-12</b><br>MCN | Site data       | Fence doesn't match the real shore (imagery offset) | Boat reaches bank or reeds inside its 'fence'  | 7 | Satellite image offset, drawing error                                 | 5 | 5 m margin / Operator watching                                                                          | 7 | 245 | 70    | R-01         | A-05    |
| <b>FM-13</b><br>HLM | RTL path        | RTL crosses an exclusion zone                       | Island or nesting area approached              | 8 | Straight-line RTL; no path planning configured (found at SDR, RID-01) | 2 | Dijkstra path planning round zones (OA_TYPE 2) with slide avoidance (HLM-D19) / Exclusion breach → stop | 2 | 32  | -     | SC-27        | -       |
| <b>FM-14</b><br>HLM | Fence           | Persistent breach doesn't stop the motors           | Boat keeps pushing at the shore                | 8 | No native mechanism (V-14)                                            | 3 | Interim HOLD from MCP and Mission Control / Breach alarm                                                | 5 | 120 | 48    | SC-28        | A-18    |
| <b>FM-15</b><br>HLM | Battery state   | Charge over-estimated                               | Late RTL; dies before home; drifts             | 7 | Wrong capacity, sensor calibration                                    | 4 | Calibration (PWR-D13) / mAh failsafe only                                                               | 3 | 84  | 42    | SC-39, L2-09 | A-06    |
| <b>FM-16</b><br>HLM | Stuck detection | Doesn't trigger in weed                             | Motors keep driving into weed; battery drains  | 7 | Boat creeps above the speed threshold                                 | 5 | Crash check (V-05) / Battery failsafe eventually                                                        | 5 | 175 | 84    | SC-05, L3-02 | A-07    |
| <b>FM-17</b><br>PRP | Thruster        | One motor or ESC dead or fouled                     | Boat circles; can't navigate or RTL            | 7 | Motor failure, weed, connector                                        | 4 | Keyed connectors, guards / None native                                                                  | 7 | 196 | 56    | SC-29        | A-08    |
| <b>FM-18</b><br>PRP | Motor wiring    | Direction or channel reversed                       | Spins or drives backwards on the first command | 9 | Build error                                                           | 4 | IF-05 mapping / First-motion check (A-03)                                                               | 2 | 72  | 54    | L2-01, SC-38 | A-03    |
| <b>FM-19</b><br>PRP | ESC             | Doesn't stop on signal loss                         | Runaway if the helm dies                       | 9 | ESC configuration                                                     | 3 | ESC failsafe setting / -                                                                                | 2 | 54  | -     | L2-10        | -       |

| ID                  | Item / function  | Failure mode                                | Effect                                             | S  | Cause                                                            | O | Controls: prevent / detect in use                            | D | RPN | After | Tests        | Actions    |
|---------------------|------------------|---------------------------------------------|----------------------------------------------------|----|------------------------------------------------------------------|---|--------------------------------------------------------------|---|-----|-------|--------------|------------|
| <b>FM-20</b><br>PRP | ESC              | Restarts at non-neutral when signal returns | Sudden jolt                                        | 6  | ESC configuration                                                | 3 | Neutral re-arm / -                                           | 3 | 54  | -     | L2-10        | -          |
| <b>FM-21</b><br>PRP | Prop guard       | Guard damaged                               | Finger injury risk at recovery                     | 10 | Impact, handling                                                 | 2 | 8 mm probe design / Checklist inspection                     | 5 | 100 | 50    | B-01         | A-09       |
| <b>FM-22</b><br>PWR | Key switch       | MOSFET fails short (rail always live)       | Latent: arming protection is software-only         | 6  | Component failure                                                | 2 | Rated part / Checklist rail test (A-10)                      | 5 | 60  | 36    | L2-02, L2-13 | A-10       |
| <b>FM-23</b><br>PWR | Battery          | Thermal runaway or fire in the box          | Fire, injury                                       | 10 | Cell damage, short, BMS fault; salvaged cells (Issue C: O 1 → 2) | 2 | BMS, fuse, hard cells, off-boat charging / None in operation | 7 | 140 | 50    | B-02, B-07   | A-11, A-20 |
| <b>FM-24</b><br>HUL | Box sealing      | Water ingress                               | Shorts: sudden stop or erratic behaviour           | 7  | Gland or lid seal                                                | 4 | Glands, dunk test / Moisture sensor → RTL                    | 3 | 84  | -     | SC-10, B-03  | -          |
| <b>FM-25</b><br>PWR | Harness          | Fuse blows or connector pulls               | Total power loss; drifts                           | 7  | Overload, vibration                                              | 2 | Rated fuse, strain relief / Heartbeat loss                   | 3 | 42  | -     | B-04         | -          |
| <b>FM-26</b><br>MCP | Mission computer | Crash or SD corruption                      | No photos; B4-B6 absent; mission still ends in RTL | 5  | Power cut, SD wear                                               | 4 | Read-only root / Mission Control sees MCP silent             | 2 | 40  | -     | SC-07, L1-08 | -          |
| <b>FM-27</b><br>MCP | Boat services    | Wrong command sent (bug)                    | Unwanted mode change; stuck in GUIDED              | 6  | Software bug                                                     | 3 | Command filter; bounded GUIDED (V-11) / Helm mode visible    | 3 | 54  | -     | SC-06, SC-30 | -          |
| <b>FM-28</b><br>MCP | UART link        | Traffic floods the UART                     | GCS heartbeats delayed → false link-loss HOLD      | 5  | Stream rates too high                                            | 3 | Rate budget (IF-04) / Latency logging                        | 3 | 45  | -     | L1-05        | -          |
| <b>FM-29</b><br>MCP | Camera           | Camera fails                                | No photos                                          | 4  | Ribbon, module                                                   | 3 | Strain relief / Health: camera_ok                            | 2 | 24  | -     | B-05         | -          |
| <b>FM-30</b><br>MCP | Clock            | Wrong time before GNSS                      | Photo and log times wrong                          | 2  | No RTC, no internet                                              | 5 | GNSS time sync / Flagged timestamps                          | 3 | 30  | -     | B-06         | -          |
| <b>FM-31</b><br>MCN | Mission Control  | Crash or power bank dies mid-mission        | No UI, no STOP; boat returns via B4                | 5  | Software, battery                                                | 4 | Power budget, autostart / Helm GCS failsafe                  | 2 | 40  | -     | SC-03, L1-06 | -          |

| ID                  | Item / function | Failure mode                           | Effect                                          | S | Cause                          | O | Controls: prevent / detect in use                 | D | RPN | After | Tests               | Actions |
|---------------------|-----------------|----------------------------------------|-------------------------------------------------|---|--------------------------------|---|---------------------------------------------------|---|-----|-------|---------------------|---------|
| <b>FM-32</b><br>MCN | Panel           | STOP button fails                      | Can't stop from the panel (web STOP remains)    | 7 | Wiring, switch                 | 2 | Quality switch / Checklist button test (A-13)     | 3 | 42  | 21    | L1-02               | A-13    |
| <b>FM-33</b><br>MCN | Panel           | GO pressed by accident                 | Mission starts before the adult is ready        | 6 | Child curiosity                | 4 | 1 s hold; adult arming / State shown and spoken   | 2 | 48  | -     | L1-02               | -       |
| <b>FM-34</b><br>MCN | Validator       | Accepts a mission crossing a boundary  | Exclusion approached; helm fence still protects | 8 | Software bug                   | 2 | 95% branch coverage / Helm fence                  | 2 | 32  | 16    | SC-31               | A-14    |
| <b>FM-35</b><br>MCN | Read-back       | Mismatch not detected                  | Uploaded mission differs from approved          | 7 | Canonical-form bug             | 2 | Contract tests / -                                | 2 | 28  | 14    | SC-32               | -       |
| <b>FM-36</b><br>MCN | Planner         | Plausible but unintended plan approved | Longer or different mission than wanted         | 4 | Adult approves without reading | 5 | Spoken summary, map / Visible on the map          | 5 | 100 | 60    | SC-33               | A-15    |
| <b>FM-37</b><br>MCN | Planner         | Odd or adversarial instruction         | Request to chase ducks etc.                     | 6 | Child phrasing, injection      | 3 | Intent vocabulary, site names only / Validator    | 3 | 54  | 24    | SC-33               | -       |
| <b>FM-38</b><br>MCN | Internet        | No internet                            | Templates only                                  | 3 | No mobile signal               | 6 | Offline templates / UI message                    | 2 | 36  | -     | P-01                | -       |
| <b>FM-39</b><br>MCN | Link            | Link flaps at the range edge           | Repeated HOLD/continue in MANUAL                | 5 | Range, geometry                | 6 | Pole antenna / RSSI display                       | 3 | 90  | -     | SC-34, L1-07        | -       |
| <b>FM-40</b><br>MCN | Link            | Commands delayed > 1 s                 | STOP late                                       | 7 | Congestion, interference       | 3 | Traffic budget / Latency logging                  | 3 | 63  | -     | SC-34, L1-03, L2-11 | -       |
| <b>FM-41</b><br>MCN | GCS identity    | Two ground stations connected          | Conflicting commands                            | 6 | QGC left running               | 3 | MC-014 procedure / -                              | 5 | 90  | 18    | SC-37               | A-16    |
| <b>FM-42</b><br>OPS | Launch          | Launched with wind away from the bank  | A dead boat drifts away                         | 7 | Procedure not followed         | 4 | Checklist / Operator                              | 5 | 140 | 70    | R-01                | A-19    |
| <b>FM-43</b><br>OPS | Other users     | Swimmer or paddler near the boat       | Collision risk (slow boat, guarded props)       | 9 | Shared water                   | 3 | OPS-003/004, low speed, guards / Operator lookout | 7 | 189 | 126   | R-01                | A-17    |

| ID                  | Item / function   | Failure mode                                  | Effect                                                                        | S  | Cause                                                                                     | O | Controls: prevent / detect in use                                                                                            | D | RPN | After | Tests              | Actions |
|---------------------|-------------------|-----------------------------------------------|-------------------------------------------------------------------------------|----|-------------------------------------------------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------------|---|-----|-------|--------------------|---------|
| <b>FM-44</b><br>OPS | Wildlife          | Nesting area approached                       | Disturbance                                                                   | 8  | Exclusions missing or too small                                                           | 2 | OPS-005 exclusions; 15 m nest stand-off enforced by the site linter (MCN-D65, CR-07); photo stops only outside it / Operator | 3 | 48  | -     | R-01, SC-26, SC-27 | -       |
| <b>FM-45</b><br>OPS | Handling          | Child handles a pod with the key in           | Prop injury risk                                                              | 10 | Supervision lapse                                                                         | 2 | Guards; adult-only arming / Key-out rule (A-09)                                                                              | 5 | 100 | 50    | B-01               | A-09    |
| <b>FM-46</b><br>MCN | Home              | Home set wrongly                              | RTL goes to the wrong place                                                   | 6  | Armed before GNSS settled                                                                 | 3 | Home shown before GO / Map                                                                                                   | 3 | 54  | -     | SC-35              | -       |
| <b>FM-48</b><br>MCN | Adult unlock      | Child learns or guesses the PIN (Issue C)     | Could arm or approve without an adult. Motors still need the magnetic key.    | 6  | PIN seen or shared                                                                        | 3 | 6 digits, lockout, never shown or spoken; the magnetic key physically gates the motors / Checklist; approval log             | 3 | 54  | -     | L1-10              | -       |
| <b>FM-47</b><br>MCN | Mission upload    | Upload interrupted (partial mission)          | Wrong mission executed                                                        | 6  | Link drop mid-transfer                                                                    | 4 | Transfer protocol / Read-back (VAL-010)                                                                                      | 2 | 48  | -     | SC-36, SC-32       | -       |
| <b>FM-49</b><br>MCP | B7 heading check  | Wind drift read as a reversed compass         | Nuisance HOLD with a wrong diagnosis; boat drifts downwind                    | 5  | Wind stronger than the boat's thrust blows it backwards at RTL start (far beyond ENV-002) | 2 | Wind limit and launch-point procedure (A-19) / Alarm names the check                                                         | 3 | 30  | -     | SC-28              | -       |
| <b>FM-50</b><br>MCP | B5 weed-shed ding | Dead motor treated as weed                    | Futile astern bursts, then HOLD with a 'stuck' alarm instead of 'motor fault' | 4  | The crash check cannot tell weed from a dead motor                                        | 4 | Ends in HOLD + alarm either way / Adult inspects on recovery                                                                 | 3 | 48  | -     | SC-29              | -       |
| <b>FM-51</b><br>MCP | Boat services     | Command sent before the helm reports the mode | Filter refuses it: the first second of each weed burst is lost                | 4  | Heartbeat (1 Hz) lags the mode change                                                     | 8 | - / Filter refusal log                                                                                                       | 2 | 64  | 8     | SC-06              | A-22    |

| ID                  | Item / function | Failure mode                                           | Effect                                                                      | S        | Cause                                                                                                         | O  | Controls: prevent / detect in use       | D | RPN        | After     | Tests | Actions |
|---------------------|-----------------|--------------------------------------------------------|-----------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------|----|-----------------------------------------|---|------------|-----------|-------|---------|
| <b>FM-52</b><br>MCP | B5 astern burst | Boat pivots instead of backing out                     | Turns into the weed; may foul the second pod                                | <b>6</b> | Negative body-frame velocity target: Rover turns round and drives forwards (specification error, ICD Issue D) | 10 | - / None before simulation              | 7 | <b>420</b> | <b>12</b> | SC-06 | A-23    |
| <b>FM-53</b><br>HLM | Modes           | Arms into MANUAL instead of HOLD                       | The boat is armed in a mode that obeys stick input; surprise for the adult  | <b>5</b> | RC mode switch overrides INITIAL_MODE (firmware defaults)                                                     | 8  | - / Mode shown at Mission Control       | 3 | <b>120</b> | <b>10</b> | SC-40 | A-24    |
| <b>FM-54</b><br>MCN | Validator       | A far-away waypoint stalls validation                  | Planning hangs for about a minute; no plan shown                            | <b>3</b> | Legs sampled every 1 m regardless of length (0° N 0° E: 5,800 km)                                             | 5  | - / Unit test found it                  | 3 | <b>45</b>  | <b>6</b>  | SC-31 | A-25    |
| <b>FM-55</b><br>MCN | Helm interface  | Parameter values lost in transfer                      | False baseline differences block arming on a good boat                      | <b>3</b> | A few of ~1,300 PARAM_VALUE messages dropped on a busy link                                                   | 6  | - / Arming refused with a list of names | 2 | <b>36</b>  | <b>6</b>  | SC-24 | A-26    |
| <b>FM-56</b><br>MCN | Session         | A boat-service stop is announced wrongly or not at all | Adult not told why the boat stopped; child hears 'stuck in weed' repeatedly | <b>5</b> | The mode change arrives before the service's reason text                                                      | 6  | - / Map shows HOLD                      | 5 | <b>150</b> | <b>20</b> | SC-43 | A-27    |
| <b>FM-57</b><br>MCN | Session         | Return home not recognised                             | No 'coming home' or 'I'm back', no auto-disarm at home                      | <b>4</b> | The final RTL item runs in AUTO; the mode never becomes RTL                                                   | 10 | - / Seen in the first end-to-end run    | 3 | <b>120</b> | <b>8</b>  | SC-41 | A-28    |

| ID                  | Item / function   | Failure mode                               | Effect                                                                                                            | S        | Cause                                                                                                                   | O | Controls: prevent / detect in use                                                                                              | D | RPN        | After     | Tests               | Actions |
|---------------------|-------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------|---|------------|-----------|---------------------|---------|
| <b>FM-58</b><br>MCP | B5 weed-shed ding | Free test misjudges the boat               | Freed boat held with a 'still stuck' alarm; or a boat with a dead motor 'freed' again and again and never stopped | <b>6</b> | Free test used ground-speed magnitude, so astern drift left by the burst counted as moving; no limit on repeat episodes | 6 | - / Found by SC-06 intermittency and MCN-D60                                                                                   | 4 | <b>144</b> | <b>36</b> | SC-06, SC-29, SC-43 | A-29    |
| <b>FM-59</b><br>HLM | RTL path          | RTL stalls short of home near an exclusion | Boat stops inside the fence, not home; adult must drive it back                                                   | <b>6</b> | Planner leg passes close to a zone and fence avoidance set to 'stop' halts the boat; or the planner finds no path       | 2 | AVOID_BEHAVE slide; OA_MARGIN_MAX > FENCE_MARGIN + corner cut (HLM-D19) / Mission Control shows RTL with no progress; operator | 3 | <b>36</b>  | -         | SC-27               | -       |
| <b>FM-60</b><br>MCN | Home              | Boat re-armed away from the launch point   | Home resets to where it was re-armed, maybe mid-lake; RTL goes there                                              | <b>7</b> | Adult disarms and re-arms on the water (e.g. after a STOP)                                                              | 3 | Arm at the jetty (CL-13); VAL-004 home sanity check before approval / Mission Control refuses the plan and says why            | 2 | <b>42</b>  | -         | SC-35, SC-09        | -       |

## 4. Results

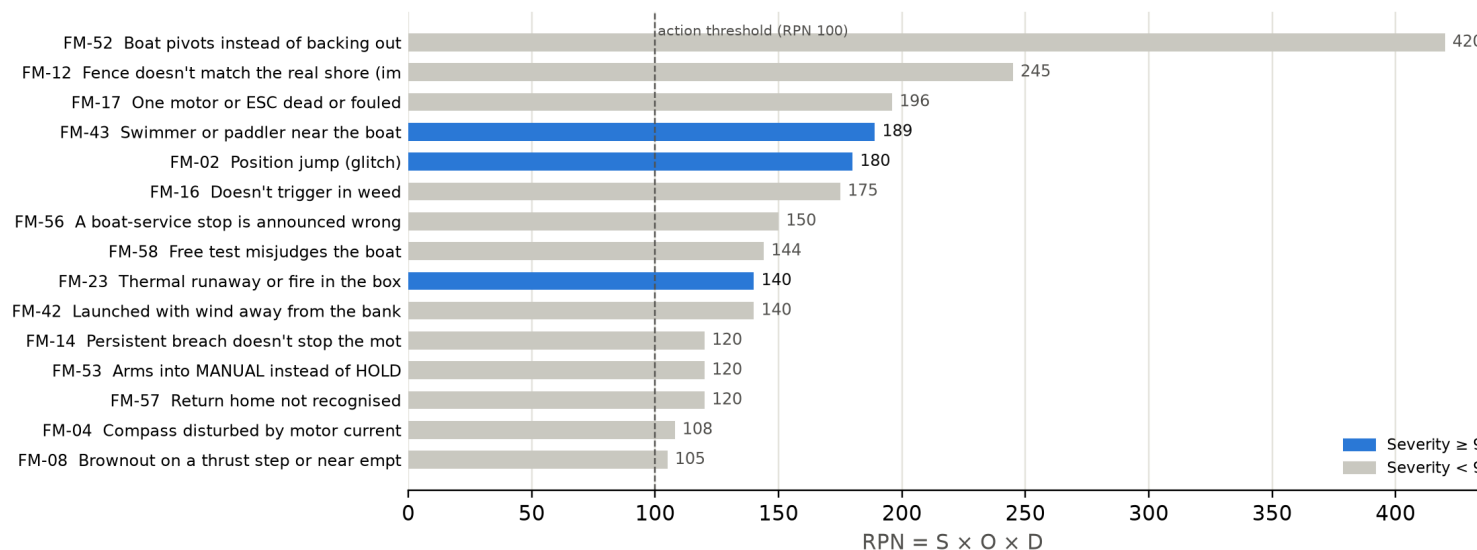


Figure 1. The 15 highest RPNs. Severity  $\geq 9$  rows need action or a direct test whatever their RPN.

Severity profile: 3 rows at S10, 9 at S9, 4 at S8; 16 rows at  $S \geq 8$ , all with tests. 25 rows meet an action rule. Those with a direct test ( $D \leq 3$ ) are covered by the test; the rest have actions.

### 4.1 What the FMEA changed

- **Three new boat-side safeguards** (actions A-03, A-07, A-08). A first-motion heading check catches a reversed compass or motor. A second stuck detector covers weed creep. A navigation-divergence watchdog catches a dead motor. All three only request HOLD, so they stay within the ADD's rule for boat-side Python.
- **Procedures matter as much as code.** The fence-vs-shore offset (A-05), the launch point for the wind (A-19), the lookout for other water users (A-17) and key-out handling (A-09) come from operational failure modes. They go into an operations manual.
- **Twenty FMEA-derived simulator scenarios (SC-20 to SC-39)** and thirteen iron-bird rig tests. The first catalogue only covered the failsafes we'd already designed.

### 4.2 After the incorporated actions

Every action is now incorporated: in the subsystem specs, the ICD or the operations manual (BOATY-OPS-001). Rows still at or above the action threshold after their actions: **FM-43** (126). FM-43 (other water users near the boat) is a severity-9 hazard with no sensor to detect it; detection relies on the lookout

procedure (OP-10), and it is an accepted residual risk (section 6). FM-02 (GNSS glitch near the fence) stays above the threshold until SC-20 runs in the simulator; Issue C wrongly listed FM-43 as the only such row. Highest RPNs now: FM-43 126, FM-39 90, FM-16 84, FM-24 84, FM-02 72, FM-04 72.

### 4.3 What the simulator changed (Issue D)

- **The worst failure mode was a specification error.** FM-52: the ICD's astern-burst command (a negative velocity target) makes Rover turn round and drive forwards into the weed. RPN 420 before the fix, the highest in this FMEA, and no planned test before simulation would have caught it.
- **Supply brown-out (FM-08) was under-rated.** The Key Component List showed the baseline flight controller's 9 V minimum is crossed by pack sag near empty (CR-04).
- **Three configuration or integration faults** (FM-51, FM-53) were found and fixed the first time the parts ran together.
- **Two new, accepted behaviours** (FM-49 gale misdiagnosis, FM-50 dead motor looks like weed) end safely in HOLD with an alarm.
- **Detection improved** where a scenario now passes (FM-05, FM-14, FM-17). FM-16 is unchanged: the helm's crash check is noise-sensitive, and B7 backs it up at 10 s.

## 5. Actions

| ID          | Action                                                                                                                                            | Owner     | Failure modes | Status                                            | Carried by                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|---------------------------------------------------|----------------------------|
| <b>A-01</b> | Confirm GNSS glitch and freeze protection parameters; add SC-20/21                                                                                | HLM       | FM-02, FM-03  | Incorporated                                      | HLM-D39                    |
| <b>A-02</b> | Measure compass interference on rig L2 before the hull layout is frozen                                                                           | HLM       | FM-04         | Incorporated                                      | HLM-D41                    |
| <b>A-03</b> | Add a first-motion heading check: in the first 10 s of any driven mode, GNSS course vs heading differs > 45° → HOLD + alarm                       | MCN / MCP | FM-05, FM-18  | Incorporated                                      | MCP-D22, MCN-D60           |
| <b>A-04</b> | Site linter: home inside the fence; site within 1 km of the configured reference; coordinate-order sanity                                         | MCN       | FM-11         | Incorporated                                      | MCN-D53                    |
| <b>A-05</b> | Fence survey procedure: walk the launch bank with a GNSS reference to check imagery offset before the first mission at a site                     | OPS       | FM-12         | Incorporated                                      | OP-01                      |
| <b>A-06</b> | Add voltage thresholds as a backstop to the mAh battery failsafe                                                                                  | HLM       | FM-15         | Incorporated                                      | HLM-D40                    |
| <b>A-07</b> | Second stuck detector in the MCP: throttle high and progress along track low for 10 s → HOLD                                                      | MCP       | FM-16         | Incorporated                                      | MCP-D23                    |
| <b>A-08</b> | Navigation-divergence watchdog in the MCP: cross-track > 10 m or heading error > 60° for 20 s → HOLD + alarm                                      | MCP       | FM-17         | Incorporated                                      | MCP-D24                    |
| <b>A-09</b> | Checklist and procedure: key out before anyone touches the boat; lift only by the handle or hoop; inspect guards                                  | OPS       | FM-21, FM-45  | Incorporated                                      | OP-05, OP-09, OP-11, CL-09 |
| <b>A-10</b> | Checklist step: key out → confirm motor rail reads 0 V on screen                                                                                  | MCN       | FM-22         | Incorporated                                      | MCN-D54                    |
| <b>A-11</b> | Box temperature sensor with alarm and RTL above 60 °C; battery inspection before each session                                                     | MCP / OPS | FM-23         | Incorporated                                      | MCP-D26, OP-04, CL-07      |
| <b>A-13</b> | Checklist button test: each crew button pressed, LED and sound confirm                                                                            | MCN       | FM-32         | Incorporated                                      | MCN-D55                    |
| <b>A-14</b> | Property-based validator tests against an independent geometry implementation                                                                     | SIM       | FM-34         | Incorporated                                      | SIM-D09                    |
| <b>A-15</b> | Plan preview shows duration and distance prominently before approval                                                                              | MCN       | FM-36         | Incorporated                                      | MCN-D56                    |
| <b>A-16</b> | C7 detects a foreign system-255 heartbeat and refuses to operate                                                                                  | MCN       | FM-41         | Incorporated                                      | MCN-D57                    |
| <b>A-17</b> | Operator is the dedicated lookout; brief 'STOP if in doubt'; no missions while people are within 50 m of the planned route                        | OPS       | FM-43         | Incorporated                                      | OP-10, CL-03, CL-17, K-06  |
| <b>A-18</b> | Resolve V-14; if there's no native mechanism, specify the interim HOLD path formally                                                              | HLM       | FM-14         | Incorporated; V-14 answered (no native mechanism) | HLM-D24, MCN-D59, MCP-D30  |
| <b>A-19</b> | Site file lists the preferred launch point per wind direction; the checklist asks 'is the wind blowing towards us?' and suggests the launch point | MCN / OPS | FM-42         | Incorporated                                      | MCN-D58, OP-06, CL-06      |

| ID          | Action                                                                                                                                                                               | Owner     | Failure modes | Status       | Carried by                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------------|--------------|---------------------------|
| <b>A-20</b> | Salvaged-cell acceptance tests and records (capacity, internal resistance, self-discharge, matching, visual)                                                                         | PWR       | FM-23         | Incorporated | PWR-D20, PWR-D21          |
| <b>A-21</b> | Flight controller that runs from 7 V (CR-04) and a motor power limit so sag near empty cannot brown out any supply                                                                   | HLM / PWR | FM-08         | Incorporated | HLM-D02, HLM-D43, PWR-D22 |
| <b>A-22</b> | Services wait until the helm reports the mode before sending mode-dependent commands                                                                                                 | MCP       | FM-51         | Incorporated | MCP-D32                   |
| <b>A-23</b> | Astern bursts by SET_ATTITUDE_TARGET thrust with zero yaw rate; the filter allows only that form                                                                                     | MCP       | FM-52         | Incorporated | MCP-D18, MCP-D19          |
| <b>A-24</b> | Boot and arm into HOLD; no RC mode switch; RC receiver ignored                                                                                                                       | HLM       | FM-53         | Incorporated | HLM-D05                   |
| <b>A-25</b> | Validator checks each leg's ends before sampling it                                                                                                                                  | MCN       | FM-54         | Incorporated | MCN-D39                   |
| <b>A-26</b> | Fetch dropped parameters again by index before comparing                                                                                                                             | MCN       | FM-55         | Incorporated | MCN-D45                   |
| <b>A-27</b> | Take the reason from the service event; announce an unexplained HOLD after 1 s; weed once per episode                                                                                | MCN       | FM-56         | Incorporated | MCN-D60                   |
| <b>A-28</b> | Follow mission progress by item sequence                                                                                                                                             | MCN       | FM-57         | Incorporated | MCN-D62                   |
| <b>A-29</b> | Free = forward speed along the heading > 0.2 m/s for 1 s within 8 s; more than 3 episodes in 2 min: HOLD + 'repeatedly stuck'; a failed mode switch is reported as 'no control'      | MCP       | FM-58         | Incorporated | MCP-D34                   |
| <b>A-30</b> | B6 holds (latched, no automatic RTL) when the helm's position jumps further than the boat could move: a sustained GNSS offset otherwise took the boat 12 m outside the fence (SC-20) | MCP       | FM-02         | Incorporated | MCP-D35                   |

A-12 is unused: it was merged into A-13 during drafting. Actions going into 'SSS-x Issue B' are carried into those specification issues next.

## 6. Residual risks accepted for Mk1

| Ref          | Residual risk and rationale                                                                                                                                                                                                                                                                                                                    |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FM-43        | Swimmer or paddler near the boat. No sensor detects people. Relies on procedure, low speed, guarded props and the operator lookout. Accepted for Mk1 with A-17.                                                                                                                                                                                |
| FM-31        | If Mission Control dies, the Wi-Fi access point dies with it, so there's no remote STOP. The boat returns by itself (B4 at 60 s). Accepted: severity 5.                                                                                                                                                                                        |
| FM-23        | Battery fire can't be detected early on board. Mitigated by hard cells, BMS, fuse and off-boat charging (occurrence 1), plus A-11. Accepted.                                                                                                                                                                                                   |
| Common cause | The fence and the failsafes both run on the flight controller, so a flight-controller logic fault is a common cause. Mitigations: a pinned stable firmware, ESC stop on signal loss, the shore physically bounds the lake, a short mission time, and operator STOP. Accepted for Mk1. A future independent guardian (SAF-005) would remove it. |

## 7. Test coverage of severe failure modes

| FM    | S  | Failure mode                              | Tests (kind)                               |
|-------|----|-------------------------------------------|--------------------------------------------|
| FM-21 | 10 | Guard damaged                             | B-01 (BENCH)                               |
| FM-23 | 10 | Thermal runaway or fire in the box        | B-02 (BENCH), B-07 (BENCH)                 |
| FM-45 | 10 | Child handles a pod with the key in       | B-01 (BENCH)                               |
| FM-02 | 9  | Position jump (glitch)                    | SC-20 (SIM), SC-13 (SIM)                   |
| FM-03 | 9  | Frozen but reported valid                 | SC-21 (SIM), L2-04 (L2)                    |
| FM-05 | 9  | Compass orientation wrong                 | SC-22 (SIM), SC-38 (SIM), L2-06 (L2)       |
| FM-09 | 9  | Parameters differ from baseline           | SC-24 (SIM)                                |
| FM-10 | 9  | Fence not loaded or disabled              | SC-25 (SIM)                                |
| FM-11 | 9  | Coordinates swapped or wrong site         | SC-26 (SIM)                                |
| FM-18 | 9  | Direction or channel reversed             | L2-01 (L2), SC-38 (SIM)                    |
| FM-19 | 9  | Doesn't stop on signal loss               | L2-10 (L2)                                 |
| FM-43 | 9  | Swimmer or paddler near the boat          | R-01 (REHEARSAL)                           |
| FM-13 | 8  | RTL crosses an exclusion zone             | SC-27 (SIM)                                |
| FM-14 | 8  | Persistent breach doesn't stop the motors | SC-28 (SIM)                                |
| FM-34 | 8  | Accepts a mission crossing a boundary     | SC-31 (SIM)                                |
| FM-44 | 8  | Nesting area approached                   | R-01 (REHEARSAL), SC-26 (SIM), SC-27 (SIM) |